Advanced Database Systems Portfolio: Kolaro Wallet

Course: MIT 8103 Advanced Database Systems, First Semester CA, 2026/2027 **Student:** Ikenna Odoh, 2026/A/MIT/0310
GitHub repository (private, shared with the assessor): https://github.com/ikenna042/advanced-database-portfolio

1. Selected organisation: Kolaro Wallet (simulated)

Kolaro Wallet is a simulated Nigerian digital wallet, invented for this portfolio; all data is synthetic. Customers hold naira, fund their wallets, send money to each other and pay merchants.

- **Users of the database:** customers (through the app), merchants, customer support, compliance officers, finance and reconciliation staff, and auditors.
- **Major data:** customers, KYC documents and reviews, wallets, merchants, ledger transactions and entries, transfers, deposits, withdrawals, reversals, idempotency keys and audit logs.
- **Key operations:** open a wallet, deposit, withdraw, transfer, reverse a failed transaction, freeze a wallet, produce a statement, audit balances against the ledger, and report volumes.

The same case study, schema and data are used in all five portfolios. Everything runs in a local Docker lab with PostgreSQL 16, MongoDB 7.0 and Redis 7.4, plus a replication cluster and a Citus cluster for Portfolio 5. Each portfolio folder has a README, its SQL or scripts, an `evidence/` folder holding the raw output of its last run, and one `run.sh` that rebuilds everything from nothing.

2. The five portfolio activities

Portfolio 1: Database design and modelling

The relational schema has 16 tables (customers, KYC, wallets, merchants, a double-entry ledger, transfers, deposits, withdrawals, reversals, idempotency keys and an audit log), normalised to BCNF in `normalisation.md`. Money is stored as whole kobo in `BIGINT`; the ledger is append-only and double-entry, with each transaction's entries netting to zero; and requests such as transfers are kept separate from posted entries, so a failed request moves no money.

Rules are enforced with primary keys, foreign keys, `UNIQUE`, expression and partial unique indexes, `CHECK` constraints, and deferred constraint triggers. **All 31 constraint tests pass**, and the ER diagrams are generated from Mermaid source.

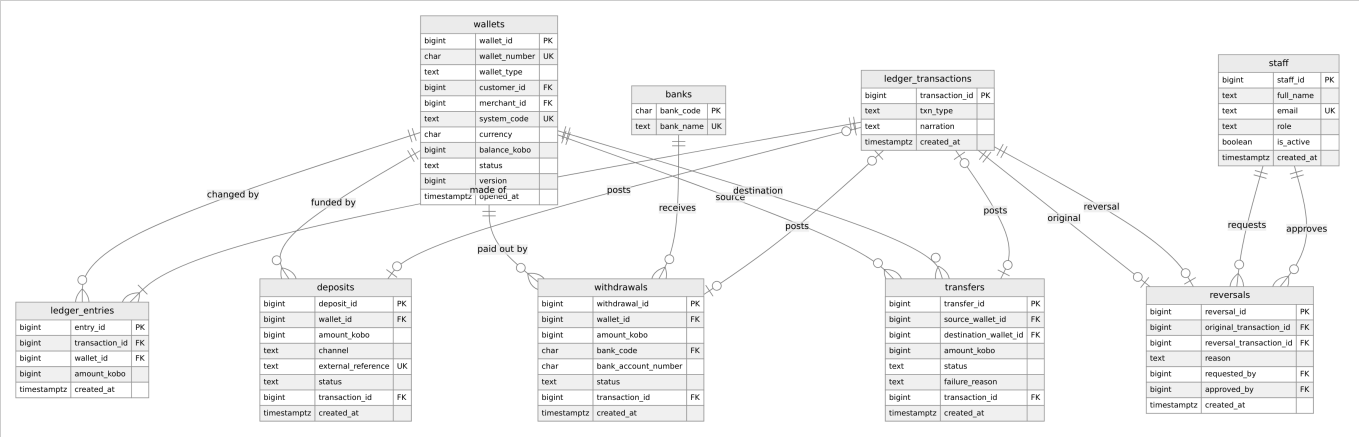


Figure 1. Portfolio 1 ER view of the ledger and money operations (full diagram in the repository).

Portfolio 2: Query processing and optimisation

A seeded generator produced a year of activity: 100,000 customers, 2.08 million ledger transactions and 4.16 million entries. Eleven queries tied to real operations were each timed five times (median), before and after indexing, using `EXPLAIN (ANALYZE, BUFFERS)`. Four further experiments compared B-tree, BRIN and partial indexes, partitioning, and the planner's estimates.

Query	Before (ms)	After (ms)	What changed
Q1b customer statement, rewritten	208.0	0.42	Covering index matching `ORDER BY … LIMIT`: 499 times faster
Q7 three-hop money trail (recursive)	437.4	0.31	An index on the recursive join column: 1,430 times faster
Q2a/Q2b customer by email	7.5	0.09	Q2a didn't use the `lower(email)` index until the query was rewritten to match it
Q8 count dormant wallets	111.3	157.6	Slower: an index can't help a query that must read most rows

Portfolio 3: Transactions and concurrency

Two-session demonstrations (driven with `dblink`) show commit, rollback, savepoints and deferred constraints; dirty reads, non-repeatable reads and phantoms at each isolation level; write skew under `REPEATABLE READ` and its prevention by

`SERIALIZABLE`; deadlock, row locks, optimistic locking and `SKIP LOCKED`; and crash recovery, where the server was killed with SIGKILL and recovered through WAL replay.

Triggers now enforce the three rules Portfolio 1 left open. A `transfer()` procedure locks both wallets in `wallet_id` order and uses idempotency keys. 17 rule tests pass. In 15-second stress runs (16 clients, 20 wallets):

Code	Isolation	Transfers per second	Money created from nothing
Naive read-then-write	Read committed	36	**2,184,302 kobo** (lost updates); 20 of 20 balances wrong
Naive read-then-write	Serializable	26	0, but 53% of transactions failed after 10 retries
`transfer()`	Read committed + ordered row locks	**1,686**	**0**, with no deadlocks and no retries

Portfolio 4: NoSQL and advanced data models

MongoDB holds four `$jsonSchema`-validated collections loaded from PostgreSQL. The counts match exactly: 100,003 customers, 882,099 statement entries and 127,410 transfers.

Customers embed their wallet and KYC history; statements use the bucket pattern (one document per wallet per month, at most 500 entries); transfers are graph edges for `$graphLookup`; audit events have flexible fields and a TTL index.

Indexing: a money-trail traversal fell from 13,050 ms to 18 ms, while an aggregation over 246,631 entries didn't improve, because unwinding them all costs more than finding them.

Single-document atomicity: 50 concurrent payment requests were sent against a daily limit. The naive version approved all 50, putting spending N50,000 over the limit. A single atomic update approved exactly 25.

Redis: idempotency claims (`SET NX`: 1 of 20 identical requests won), fixed-window and Lua sliding-window rate limits, sessions, cache-aside with invalidation, and a capped activity feed.

Suitability: the ledger stays relational. MongoDB serves read models and Redis holds short-lived state (polyglot persistence). A Cassandra table was designed but not run.

Portfolio 5: Distributed and cloud databases

Replication: three PostgreSQL nodes with a synchronous and an asynchronous standby. The primary was killed while the application was sending transfers.

The synchronous standby held all 264 confirmed transfers, so nothing was lost; the lagging asynchronous standby held none of them. Recovery took 10.6 s, 9.7 s of which was failure detection. Waiting for a standby 10 ms away raised commit latency from 4.1 ms to 20.0 ms. With the only synchronous standby down, commits blocked; quorum commit (`ANY 1 of 2`) kept them flowing. A restarted old primary caused real **split brain**: it committed a transfer that reused an ID already issued by the new primary, and `pg_rewind` then turned it back into a standby.

Sharding (Citus, the 4.2 million-entry data set, sharded on `wallet_id`):

Statements became single-shard router queries. A fan-out aggregation ran in 1.49 s, against 2.65 s on one server, while a join on a column that isn't the shard key took 24.4 s, against 0.89 s. 97% of real transfers span shards and commit with two-phase commit (`PREPARE TRANSACTION`, then `COMMIT PREPARED`). A failed worker made its shards unavailable rather than returning wrong answers. Adding a worker moved shards in 143 s while 18,373 transfers committed with 0 failures.

The README relates each result to CAP and PACELC and proposes the cloud deployment in Figure 2.

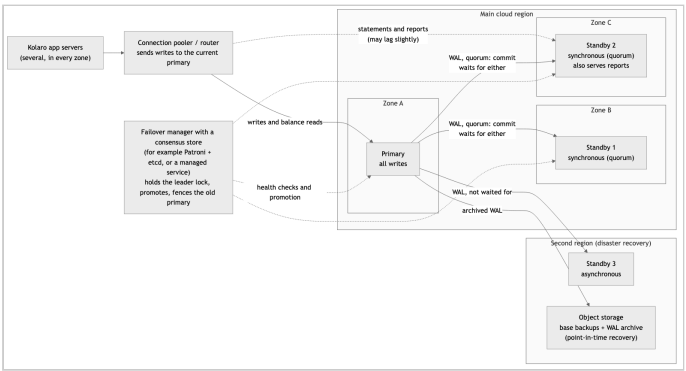


Figure 2. Proposed deployment (Portfolio 5): a quorum of synchronous standbys across zones, an asynchronous disaster-recovery standby in a second region, and a WAL archive.

Sample evidence (extract from `portfolio-5-distributed-cloud/evidence/r5_failover.txt`)

```
Every transfer node1 confirmed before it died, looked up on the other nodes:
[node2] ... confirmed_by_node1 | missing_here        [node3] ... confirmed_by_node1 | missing_here
                          264 |            0                                    264 |          264
Last confirmation from node1: +0.09 s   node2 promoted: +10.26 s   First confirmation from node2: +10.64 s
```

Every figure in this document comes from these evidence files, which the scripts in the repository regenerate. The repository's commits show the portfolio developing from 4 to 10 November 2026: the lab and Portfolio 1 first, then one commit per portfolio.

3. Individual technical reflection

Design choices. I chose a wallet because money gives every topic a clear test of correctness: the ledger must always balance, and no transfer may be lost or posted twice. Portfolio 1's double-entry, append-only design made that testable, and each later portfolio was judged against it: a fast query must return the same answer, and a concurrent or distributed run must leave the trial balance at zero. I kept the ledger relational throughout. Portfolio 4 showed that MongoDB and Redis suit the read models and short-lived state, but not the multi-row transactions that move money.

Challenges. Several results only became trustworthy after I questioned them.

- An early test run reported success with zero tests, because a rollback also erased the results table.
- My first lost-update demonstration failed for an unexpected reason: the CHECK constraint fired before the second session even waited for the lock.
- In Portfolio 5, failover detection first looked slow because of Docker's DNS, not PostgreSQL. Using fixed IP addresses fixed the measurement.
- A Citus placement view returned no rows because each worker gives the same table a different internal OID.
- During one stress run, PostgreSQL raised an internal error (`new multixact has more than one updating member`). I recorded it and matched it to an open upstream report rather than claiming to diagnose it.

Lessons learned.

- An index helps only when the query is written to use it, and can make a query slower (Q2a and Q8).
- Correctness under concurrency comes from the right locks, not just a higher isolation level: ordered row locks were both correct and 47 times faster than the naive code.
- "Synchronous" replication means durable on the standby, not visible there.
- In a sharded system, the choice of shard key decides which queries scale. Sharding `ledger_transactions` separately made almost every transfer a distributed transaction, a flaw I would correct by co-locating it with the source wallet.
- The biggest lesson is to trust measurements over assumptions, and to record surprising results honestly, including those that didn't improve.

4. AI use declaration

Tool: Claude Code (Anthropic), using the Claude Opus 5.5 model.

Used for: planning the portfolio; writing the Docker configuration, SQL, MongoDB, Redis and shell scripts, tests and READMEs; running the scripts to capture evidence; and drafting this document.

How the output was reviewed and verified:

Every claim is backed by output from a real run, saved in each `evidence/` folder and reproducible with `run.sh`; no result was typed in by hand. Built-in checks make the scripts fail when results are wrong: constraint and rule tests, trial-balance and balance-audit queries, and PostgreSQL/MongoDB count comparisons. Figures in the READMEs were cross-checked against the evidence files, and errors found this way were corrected, for example a wrongly reported entry count in Portfolio 4. I studied each portfolio's concepts and results from my revision notes, so that I can explain and rerun any part of the work.

5. References

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Tools: Docker/OrbStack, pgbench, dblink, mermaid-cli.